



OPEN Awareness, usage, and perspectives on ChatGPT in dental education among graduate and undergraduate students

Raahim Salman Abdul Ghaffar¹, Raiqa Malik², Muqaddas Amjad², Ruba Nadeem², Muhammad Saad Admani², Anas Sanaullah², Sara Altamash³, Naseer Ahmed², Artak Heboyar^{4,5,6}✉ & Rizwan Jouhar⁷✉

To assess the knowledge, attitudes, and practices (KAP) related to ChatGPT among undergraduate students and graduates in Karachi, Pakistan, and to analyze an association with sociodemographic factors. A cross-sectional study was conducted over three months using a self-administered, structured questionnaire distributed via online platforms (WhatsApp, and emails). The questionnaire included four sections: demographics, knowledge, attitude, and practice related to ChatGPT. A non-probability convenience sampling technique was employed. Data were analyzed using SPSS version 26. Descriptive statistics summarized participant characteristics. Multiple linear regression was used to identify associations between KAP outcomes and sociodemographic variables. A p -value ≤ 0.05 was considered statistically significant. A total of 347 participants responded, of whom 74.6% were female and 76.4% reported awareness of ChatGPT. Age, designation, and years of experience were significantly associated with knowledge ($p = 0.012$, 0.001 , and 0.001 , respectively). Attitude was significantly influenced by age and gender ($p = 0.03$ and 0.04 , respectively), while only gender showed a significant association with perception ($p = 0.021$). Most respondents considered ChatGPT moderately accurate (54.8%) and acknowledged its usefulness in education and research. Ethical concerns and uncertainty about ChatGPT replacing human roles were also noted. The majority of dental students and professionals in Karachi are aware of ChatGPT and hold a cautiously optimistic view of its role in dental education. While attitudes are generally positive, concerns remain regarding ethical implications and clinical applicability. Structured AI training and clear implementation guidelines are needed to support its effective integration in dental practice.

Keywords Dental students, Education, Chat-GPT, Artificial intelligence, AI chatbot

Artificial intelligence (AI) refers to the intelligence displayed by machines or software, due to its mimicking cognitive function for real-world problems and developing systems that think like people¹. Artificial Intelligence has had a profound impact on various aspects of life, addressing complex challenges across scientific, engineering, business, medical, meteorological, and media domains^{2–6}. The advancement of AI technology has stirred both aspirations and concerns, particularly within the healthcare sector, including the detection of radiographs, viewing histopathological slides, detecting diabetic eye diseases, and many more applications in medicine⁷. Artificial intelligence also has a great impact on the field of dentistry⁸. Artificial Intelligence methodologies have great capability to detect and diagnose the lesions of the oral cavity, which may usually go undetected by the human eye, therefore making their way towards dental practice⁹. The potential of advanced A.I. in orthodontics

¹Department of Paediatrics & Child Health, Aga Khan University, Karachi, Pakistan. ²Department of Prosthodontics, Altamash Institute of Dental Medicine, Karachi 75500, Pakistan. ³Department of Orthodontics, Altamash Institute of Dental Medicine, Karachi 75500, Pakistan. ⁴Department of Research Analytics, Saveetha Dental College and Hospitals, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai 600 077, India. ⁵Department of Prosthodontics, Faculty of Stomatology, Yerevan State Medical University after Mkhitar Heratsi, Str. Koryun 2, 0025 Yerevan, Armenia. ⁶Department of Prosthodontics, School of Dentistry, Tehran University of Medical Sciences, North Karegar St, Tehran, Iran. ⁷Department of Restorative Dental Sciences, College of Dentistry, King Faisal University, 31982, Al Hofuf, Al-Ahsa, Saudi Arabia. ✉email: a_heboyar@farabi.tums.ac.ir; rjouhar@kfu.edu.sa

has been explored particularly for applications such as CBCT diagnosis and assessment, evaluation, and treatment progress, ensuring the solidity of outcomes during the follow-up stage¹⁰. Using artificial intelligence (AI) in dental practice can face challenges like getting data, interpreting it, and ethical concerns. Still, AI is considered a worthy addition to dental procedures⁸.

In November 2022, ‘Open AI,’ a notable artificial intelligence company, unveiled a widely recognized chatbot named ‘ChatGPT’¹¹. This versatile chatbot has multifaceted applications, ranging from educational and entertainment purposes to predicting queries, drafting applications, conducting research, managing schedules, making appointments, and assisting in debugging codes¹². Notably, it has found valuable applications in the healthcare sector, providing medical guidance and information, such as responding to medical inquiries, offering recommendations for drugs and treatments, and aiding in the diagnostic process¹³. Lately, ChatGPT has played an essential role in creating discharge summaries, which can drastically reduce the workload of documentation in the healthcare sector¹⁴. Furthermore, ChatGPT could enhance service efficiency by maximizing clinical output and reducing work costs. ChatGPT has the potential to play a major role in breast cancer screening, evaluating pain and identifying necessary imaging steps¹⁵.

ChatGPT possesses the potential to significantly bolster clinical applications and research within the field of dentistry. It can also be used as an educational tool for students and professionals in the field of dentistry by offering an immense amount of knowledge and information¹⁶. Large Language Model AI Chatbot has the capability of strengthening clinical application and research in dentistry. It is positioned to enhance dental diagnosis and treatment planning by surmounting dental practitioners in the accuracy and efficiency of diagnostic procedures, therefore decreasing the occurrence of missed findings and preventing overtreatment¹⁷. AI algorithms can be trained to evaluate CBCT images for dental conditions by examining the alignment, location of teeth, and assessing bone quality for placing implants, thereby resulting in AI-assisted image processing and analysis. ChatGPT can offer patients accurate and reliable information tailored to their specific dental health and hygienic needs¹⁸. ChatGPT can provide prompt feedback, address doubts, and provide additional resources to readers.

As artificial intelligence (AI) continues to evolve, tools like ChatGPT are gaining significant attention across various sectors. Globally, individuals are leveraging ChatGPT for diverse applications, ranging from education to healthcare. In Pakistan, recent studies have highlighted a growing awareness and utilization of ChatGPT among the general population. One such study reported that approximately 76% of respondents were aware of ChatGPT, with 30% of this group comprising healthcare providers¹⁹.

Building upon these findings, the current research focuses specifically on the dental community in Pakistan. It aims to assess the level of awareness, attitudes, and practices related to ChatGPT among dental professionals and students. Understanding how this AI tool influences their daily professional routines and their perspectives on its future applications is crucial. Given the potential of ChatGPT to transform educational methodologies, clinical decision-making, and patient communication, this study seeks to provide insights into its current impact and future implications within the field of dentistry in Pakistan.

The objective of the study

The objective of the study is to assess the knowledge, attitudes, and practices (KAP) related to ChatGPT among undergraduate and graduate dental students and professionals in Karachi, Pakistan, and to analyze an association with sociodemographic factors.

Methodology

A questionnaire-based cross-sectional study was conducted over a period of three months across various dental institutes in Karachi, Pakistan. The target population included:

- Students currently enrolled in Bachelor of Dental Surgery (BDS) programs.
- Dental graduates holding a BDS degree.
- Postgraduate students or dentists pursuing further professional education after obtaining their BDS degree.

Ethical approval for the study was obtained from the institutional review board. The study was conducted in accordance with the Declaration of Helsinki and its 2013 revision. A non-probability convenience sampling technique was employed. The questionnaire was disseminated via internet-based platforms, including WhatsApp, and email using generalized invitations to maximize reach. Participation was voluntary and anonymous. No names or identifying information were collected, and all data were stored securely on a password-protected device accessible only to the core research team.

The sample size was determined using the RAOSoft sample size calculator. Based on a confidence interval of 95%, a margin of error of 5%, and an assumed response distribution of 65%, the required sample size was calculated to be 347 participants.

A self-administered questionnaire was developed using Google forms and structured into four sections. Demographics include age, gender, professional status, and work sector. Participants were also asked to indicate their years of clinical experience after receiving their BDS degree. This variable was categorized into the following groups: 0 years (current students), 1–3 years, and more than 3 years. Knowledge-based items to assess awareness and understanding of ChatGPT and items specific to assess the Attitudes i.e., participants’ opinions on ChatGPT’s role in dentistry. Additionally, the questionnaire also had a section to collect data about practices of the participants or usage patterns of ChatGPT in academic or clinical settings. (Supplementary file 1).

The questionnaire underwent Pre-testing for face and content validation on 30 participants once, including heads of departments (HODs), assistant professors, research specialists, and research managers. Their feedback

Gender	No	Yes	Total Participants
Female	66 (25.6%)	192 (74.4%)	258 (100%)
Male	16 (18%)	73 (82%)	89 (100%)
Total	82 (23.6%)	265 (76.4%)	347 (100%)

Table 1. Gender distribution and familiarity with ChatGPT.

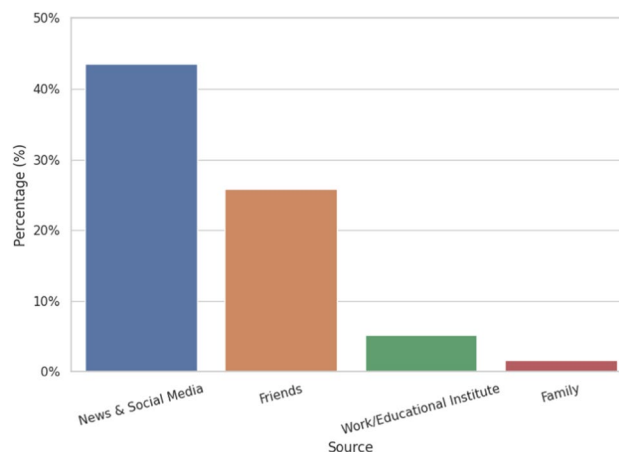


Fig. 1. Source of information regarding ChatGPT.

was used to refine the questionnaire in terms of clarity, structure, and relevance. To assess internal consistency, the Cronbach's alpha coefficient was calculated, yielding a value of $\alpha = 0.71$, indicating acceptable reliability.

Statistical analysis

Data was analyzed using SPSS software version 26. Descriptive statistics, including frequencies and means, were used to summarize the demographic characteristics of the participants. Multiple linear regression analysis was performed to evaluate the association between participants' knowledge, attitude, and perception of ChatGPT and their sociodemographic variables. For the purpose of regression analysis, all binary response variables (e.g., 'Yes'/'No') were coded as 1 and 2, respectively, to convert nominal data into numeric format suitable for analysis. A p -value of ≤ 0.05 was considered statistically significant.

Results

The majority of participants were female (74.6%, $n = 258$), among whom 74.4% reported familiarity with ChatGPT. Among male participants ($n = 89$), an even higher proportion (82%) were familiar with the platform. Overall, 76.4% of all participants ($n = 347$) reported awareness and use of ChatGPT (Table 1). In this study most of the participants belonged to the age group of 18–25 (78%), while (19.6%) were 26–35 years whereas, the remaining (2.3%) belonged to the age range of 36–50 years respectively. The study included 153 (44.1%) dental students, 140 (40.3%) general dentists, 47 (13.5%) postgraduate students, and seven (2%) consultants. In terms of clinical experience, dental students had no professional experience, general dentists and postgraduates primarily had 1–3 years of experience, while consultants typically had more than 3 years of clinical experience. Seventy (20.2%) participants belonged to the public sector, 236 (68%) to the private sector, and 41 (11.8%) to both sectors.

News and social media emerged as the most common source of information about ChatGPT, cited by over 40% of respondents. Friends accounted for about a quarter, while fewer participants learned about ChatGPT through their workplace or educational institutions. Family was the least cited source, indicating minimal influence from home environments on ChatGPT awareness (Fig. 1).

A significant proportion of participants, more than 45%, believed ChatGPT would have a positive effect on future learning and creativity. Conversely, around 20% perceived a negative impact, while a smaller segment (10%) expressed a neutral approach. This indicates a largely optimistic outlook on the educational and creative potential of ChatGPT (Fig. 2).

Among those aware of ChatGPT ($n = 265$), about one-fourth (26%) had sought external help to learn its usage, while just over half (49.9%) had not. Routine use was reported by 25.6% of respondents. A majority found it easier to use compared to other sources (64%) and considered it helpful (62.8%). However, only 23.3% reported prior use of other AI tools during their dental training (Table 2).

The attitude of participants varied across usage contexts. ChatGPT was most commonly used as an educational tool (47%), followed by research assistance (36.3%) and answering dental questions (32.3%). A relatively smaller number of participants reported using it for administrative tasks (21%) or patient counseling

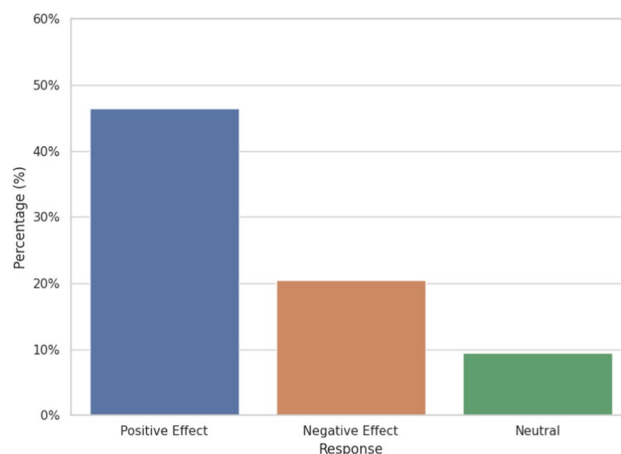


Fig. 2. The impact of ChatGPT on learning and creative performance.

S.no	Items	Yes (%)	No (%)
1	Have you ever taken help from some external source to learn how to use chat GPT?	92 (26%)	173 (49.9%)
2	Do you often use chat GPT in your daily routine?	89 (25.6%)	176 (50.7%)
3	Is it easy to use chat GPT as compared to other sources?	222 (64%)	43 (12.4)
4	Was it helpful to use Chat-GPT	218 (62.8%)	47 (13.5%)
5	Have you ever used A.I other than Chat-GPT in your dental journey	81 (23.3%)	184 (53.3%)

Table 2. The percentage distribution of knowledge among the participants regarding ChatGPT. (23.6%, who are not aware of chat-GPT were excluded).

S.no	Items	Yes (%)	No (%)
1	Do you use ChatGPT for answering dental related questions	112 (32.3%)	153 (44.1%)
2	Do you use ChatGPT for patient counseling	26 (7.5%)	239 (68.9%)
3	Do you use ChatGPT for research assistance	126 (36.3%)	139 (40.1%)
4	Do you use ChatGPT for educational tool	163 (47.0%)	102 (29.4%)
5	Do you use ChatGPT for administrative task	73 (21.0%)	192 (55.3%)

Table 3. The percentage distribution of attitude among the participants regarding ChatGPT Participants and Knowledge (n = 265) (23.6%, who are not aware of chat-GPT were excluded).

(7.5%), indicating limited clinical or managerial application among dental students and professionals (Table 3). Although the questionnaire did not include direct measures of clinical or managerial competence, responses to items such as “Use of ChatGPT for patient counseling” and “Use for administrative tasks” were interpreted as indicative of participants’ perceptions of ChatGPT’s potential in these domains.

Moreover, half of the participants (50.1%) viewed ChatGPT as a superior knowledge source. Nearly half (48.7%) believed it could replace certain academic roles in the future. Opinions on the ethical use of AI were divided almost evenly. Over half (53.3%) expressed concern that AI use may impact future dental professionals’ skills, although fewer (33.1%) anticipated AI overtaking dentists’ roles. Nonetheless, (31.4%) supported restrictions or bans on ChatGPT use in dental education (Table 4).

The regression analysis revealed that professional designation and years of experience were significant predictors of knowledge about ChatGPT ($p=0.001$), while gender significantly influenced both attitude ($p=0.04$) and perception ($p=0.021$). Age showed a borderline significance in influencing attitude ($p=0.03$) but not knowledge or perception. Other variables, such as sector, were not significantly associated with any of the outcomes (Table 5).

Discussion

The rapid evolution of artificial intelligence in healthcare, understanding how emerging tools like ChatGPT are perceived and utilized by future dental professionals is both timely and critical¹⁴. This study aimed to bridge that knowledge gap by exploring the awareness, attitudes, and practices related to ChatGPT among dental students and practitioners in Karachi, Pakistan.

S.no	items	Yes (%)	No (%)
1	Do you think chat GPT is a better source of knowledge than others?	174 (50.1%)	91 (26.2%)
2	In the future, do you think chat GPT will replace most of the low scale academic jobs such as demonstrators, writers and tutors?	169 (48.7%)	96 (27.7%)
3	Do you think it is ethically right to use A.I. in the field of healthcare and medicine?	133 (38.3%)	132 (38.0%)
4	Will the use of chat GPT affect the abilities and the performance of our future dental professionals?	185 (53.3%)	80 (23.1%)
5	Do you think the future or upgraded versions of Artificial Intelligence might take over dentists in the healthcare profession?	115 (33.1%)	150 (43.2%)
6	Should our dental educational system ban or have conditional restrictions on the use of chat GPT?	109 (31.4%)	156 (45.0%)

Table 4. The percentage distribution of perception among the participants regarding ChatGPT (23.6%, who are not aware of chat-GPT are excluded).

Variable	Knowledge				Attitude				Perception			
	St.Error	R-square	F-change	p-value	St.Error	R-square	F-change	p-value	St.Error	R-square	F-change	p-value
Age	0.466	0.005	3.007	0.012	0.469	0.45	2.44	0.03	0.474	0.025	1.11	0.355
Gender	0.446	0.027	1.442	0.21	0.442	0.43	2.35	0.04	0.44	0.055	2.483	0.021
Designation	0.737	0.078	4.357	0.001	0.75	0.38	2.06	0.07	0.76	0.017	0.76	0.6
Sector	0.702	0.006	0.318	0.9	0.696	0.25	1.3	0.263	0.701	0.014	0.59	0.73
Years of experience	0.487	0.089	5.036	0.001	0.505	0.2	1.07	0.377	0.502	0.035	1.5	0.16

Table 5. Regression analysis for knowledge, attitude, and perception showing predictors of knowledge, attitude, and perception toward ChatGPT among dental students and professionals.

Our findings indicate that over 75% of participants were aware of ChatGPT; however, its application in clinical practice remains limited. This trend aligns with a study conducted in Islamabad, where 61% of medical and dental students had utilized ChatGPT primarily for academic purposes, such as case-based and problem-based learning²⁰. Similarly, research from Rawalpindi highlighted that while students recognized ChatGPT’s potential, its usage was predominantly for academic assistance rather than clinical application²¹.

The user-friendly nature of ChatGPT was acknowledged by participants, with many finding it accessible through various platforms, notably social media. This aligns with studies indicating that social media serves as a primary conduit for disseminating information about AI tools among students^{22,23}. Regarding the perceived accuracy of ChatGPT, most participants considered its responses moderately accurate. This perception is consistent with findings from a study in Saudi Arabia, where dental students expressed moderate confidence in AI tools, emphasizing the need for enhanced AI literacy and training²⁴. It is noteworthy that this study did not directly assess clinical or managerial skills; rather, the interpretations were based on participants’ reported contexts of ChatGPT use, such as patient counselling and administrative assistance.

When exploring the potential of ChatGPT to replace certain academic roles, nearly half of the participants believed it could substitute junior faculty positions. This sentiment resonates with international discussions on AI’s role in education, where concerns about job displacement coexist with recognition of AI’s supportive capabilities²⁵.

Ethical considerations emerged as a significant concern. Participants were divided on the appropriateness of using AI in healthcare, reflecting global debates on the ethical implications of AI integration in clinical settings. A study by Wei Q et al. highlighted potential ethical challenges from legal, humanistic, algorithmic, and informational perspectives, emphasizing the need for clear rules and legal boundaries to properly allocate liability and protect users²⁶.

The reliability of ChatGPT’s responses has been tested in various scenarios, with developers presenting both simple commands and more complex, profession-related queries to assess its functionality. In our study, 54.8% of participants suggested the software was moderately accurate. The findings are well supported by other studies that indicated satisfactory performance of ChatGPT when handling complex commands^{27,28}.

Despite the intelligent capabilities of ChatGPT, some participants remained unconvinced about its ability to independently perform tasks traditionally conducted by humans. Concerns were raised about the ethical acceptability of deploying such software in the healthcare industry. In our study, 38.3% of participants agreed that it was ethically acceptable to use ChatGPT in healthcare, while a similar proportion disagreed. This uncertainty arises due to accountability dilemmas, particularly if the software provides incorrect outputs leading to adverse patient outcomes. Establishing responsibility and taking appropriate medicolegal action in such situations could be challenging.

Furthermore, the study by Alammari et al.²⁹ in Saudi Arabia revealed that 42.5% of medical students were concerned about AI-induced job displacement, and 34.4% believed AI might eventually replace traditional doctor roles. Despite these concerns, there was a unanimous recognition of the potential for AI to collaborate with human healthcare professionals, indicating a shift towards a synergistic model in clinical practice. In contrast, our study involving dental professionals and students in Karachi, Pakistan, found that 53.3% believed ChatGPT could affect the performance and abilities of dental health providers, while 43.2% did not believe it would replace dentists in their professional roles.

These findings underscore the importance of integrating AI education into medical and dental curricula to prepare students for the evolving technological landscape. Addressing ethical concerns and promoting collaborative models can alleviate fears about job displacement. Continuous professional development through workshops and training sessions can keep current practitioners informed about AI advancements and applications. By implementing these strategies, educational institutions can better equip future healthcare professionals to navigate and embrace the integration of AI in their fields.

Limitations

This study was conducted solely in Karachi, Pakistan, which limits the generalizability of the findings. A more comprehensive approach would involve conducting similar studies across different regions or countries to better assess the knowledge, attitudes, and practices related to ChatGPT on an international scale. Since ChatGPT is a relatively new and rapidly evolving platform, its ongoing updates may significantly influence user perceptions and usability over time. As a cross-sectional study, this research captures a snapshot of participant opinions at a single point in time and does not account for potential changes in perception as the software continues to develop. A longitudinal study would be more appropriate to assess how attitudes, knowledge, and practices evolve over time.

While a pre-test of the sample was conducted to evaluate face and content validity, it involved only academic experts. Including members of the target population (undergraduate and postgraduate dental students) in this pilot phase would have enhanced the representativeness and reliability of measures such as the Cronbach's alpha. One limitation of this study is the combination of undergraduate and graduate dental students within a single convenience sample, without stratified analysis. While this approach allowed for a broader perspective on ChatGPT usage across different educational levels, it may have obscured group-specific differences. Future studies should consider stratified sampling or separate analysis of educational subgroups to provide more targeted insights. Additionally, the study focused exclusively on individuals from a dental background. This may include participants who are less familiar with or less supportive of AI technologies. Moreover, younger individuals who are not currently enrolled in a dental program were excluded from the study. Including a broader age range and educational backgrounds might provide more diverse perspectives and strengthen the findings. Although years of experience was included as a variable to assess the impact of professional exposure on ChatGPT familiarity, we acknowledge that the majority of our sample consisted of students and early-career professionals. Therefore, the influence of this variable may be limited and not generalizable to experienced clinicians. Future studies should include a more diverse range of clinical experience levels to better evaluate this factor.

Despite these limitations, this study provides valuable preliminary insights into the knowledge, attitudes, and practices regarding the use of AI software like ChatGPT among dental students and graduates. It sets the foundation for future, more extensive research exploring the integration of AI technologies in healthcare education and practice.

Conclusion

This study highlights a generally positive awareness and acceptance of ChatGPT among dental students and professionals, with the majority recognizing its potential educational and research utility. While knowledge about the platform was influenced by professional designation and years of experience, gender played a notable role in shaping attitudes and perceptions. Despite concerns regarding ethical implications and job displacement, a substantial proportion of participants believed in the constructive role of ChatGPT in enhancing learning and creative performance. These findings highlight the need to integrate AI literacy into dental education while also fostering critical discourse on its ethical and professional implications.

Data availability

The datasets used and analysed during the current study are available from the corresponding author on a reasonable request.

Received: 26 February 2025; Accepted: 22 December 2025

Published online: 25 December 2025

References

- Holzinger, A., Langs, G., Denk, H., Zatloukal, K. & Müller, H. Causability and explainability of artificial intelligence in medicine. *Wiley Interdiscip. Rev. Data Min. Knowl. Discov.* **9**(4), e1312 (2019).
- Loureiro, S. M. C., Guerreiro, J. & Tussyadiah, I. Artificial intelligence in business: State of the art and future research agenda. *J. Bus. Res.* **129**, 911–926 (2021).
- Mintz, Y. & Brodie, R. Introduction to artificial intelligence in medicine. *Minim. Invasive Ther. Allied Technol.* **28**(2), 73–81 (2019).
- Ouchchy, L., Coin, A. & Dubljević, V. AI in the headlines: The portrayal of the ethical issues of artificial intelligence in the media. *AI Soc.* **35**, 927–936 (2020).
- Pan, Y. & Zhang, L. Roles of artificial intelligence in construction engineering and management: A critical review and future trends. *Autom. Constr.* **122**, 103517 (2021).
- Raschka, S., Patterson, J. & Nolet, C. Machine learning in python: Main developments and technology trends in data science, machine learning, and artificial intelligence. *Information* **11**(4), 193 (2020).
- Kulkarni, S., Seneviratne, N., Baig, M. S. & Khan, A. H. A. Artificial intelligence in medicine: where are we now?. *Acad. Radiol.* **27**(1), 62–70 (2020).
- Fa, S., Samek, W. & Krois, J. Artificial intelligence in dentistry: Chances and challenges. *J. Dent. Res.* **99**(7), 769–774 (2020).
- Gomes, R. F. T. et al. Use of artificial intelligence in the classification of elementary oral lesions from clinical images. *Int. J. Environ. Res. Public Health* **20**(5), 3894 (2023).
- Monill-González, A., Rovira-Calatayud, L., d'Oliveira, N. G. & Ustrell-Torrent, J. M. Artificial intelligence in orthodontics: Where are we now? A scoping review. *Orthod. Craniofac. Res.* **24**, 6–15 (2021).

11. Lo, C. K. What is the impact of ChatGPT on education? A rapid review of the literature. *Educ. Sci.* **13**(4), 410 (2023).
12. Wu, T. et al. A brief overview of ChatGPT: The history, status quo and potential future development. *IEEE/CAA J. Autom. Sin.* **10**(5), 1122–1136 (2023).
13. Huang, X. et al. Evaluating the performance of ChatGPT in clinical pharmacy: A comparative study of ChatGPT and clinical pharmacists. *Br. J. Clin. Pharmacol.* **90**, 232 (2023).
14. Singh, S., Djalilian, A. & Ali, M. J. ChatGPT and ophthalmology: Exploring its potential with discharge summaries and operative notes. In *Seminars in Ophthalmology* (Taylor & Francis, 2023).
15. Sorin, V. et al. Large language model (ChatGPT) as a support tool for breast tumor board. *NPJ Breast Cancer* **9**(1), 44 (2023).
16. Ali, K., Barhom, N., Tamimi, F. & Duggal, M. ChatGPT—A double-edged sword for healthcare education? Implications for assessments of dental students. *Eur. J. Dent. Educ.* **28**, 206 (2023).
17. Huang, H. et al. ChatGPT for shaping the future of dentistry: The potential of multi-modal large language model. *Int. J. Oral Sci.* **15**(1), 29 (2023).
18. Bagde, H., Dhopte, A., Alam, M. K. & Basri, R., A systematic review and meta-analysis on ChatGPT and its utilization in medical and dental research. *Heliyon* (2023).
19. Bodani, N. et al. Knowledge, attitude, and practices of general population toward utilizing ChatGPT: A cross-sectional study. *SAGE Open* **13**(4), 21582440231211080 (2023).
20. Qamar, K. et al. Exploring the adoption and utilisation of ChatGPT in everyday academic practices among medical and dental students of Rawalpindi and Islamabad: A study on the impact and perceived effectiveness. *JPMA J. Pak. Med. Assoc.* **74**(12), 2101–2106 (2024).
21. Jan, H., Fatima, A., Ahmad, M. B., Arshad, A. & Abideen, M. Z. Understanding student engagement and satisfaction with ChatGPT in dental education: A qualitative exploration. *J. Rawalpindi Med. Coll.* **28**(3) (2024).
22. Khan, N. S. et al. Use of artificial intelligence tools among medical and dental students at Ayub medical college, Abbottabad. *Pak. J. Physiol.* **21**(1), 92–95 (2025).
23. Kavadella, A., Da Silva, M. A., Kaklamanos, E. G., Stamatopoulos, V. & Giannakopoulos, K. Evaluation of ChatGPT's real-life implementation in undergraduate dental education: mixed methods study. *JMIR Med. Educ.* **10**(1), e51344 (2024).
24. Alsayed, A. A. et al. Assessing the quality of AI information from ChatGPT regarding oral surgery, preventive dentistry, and oral cancer: An exploration study. *Saudi Dent. J.* **36**(11), 1483–1489 (2024).
25. George, A. S., George, A. H. & Martin, A. G. ChatGPT and the future of work: A comprehensive analysis of AI's impact on jobs and employment. *Partn. Univers. Int. Innov. J.* **1**(3), 154–186 (2023).
26. Wei, Q. et al. Evaluation of ChatGPT-generated medical responses: a systematic review and meta-analysis. *J. Biomed. Inform.* **8**, 104620 (2024).
27. Johnson, D., Goodman, R., Patrinely, J., Stone, C., Zimmerman, E., Donald, R., Chang, S., Berkowitz, S., Finn, A., Jahangir, E. & Scoville, E. Assessing the accuracy and reliability of AI-generated medical responses: An evaluation of the Chat-GPT model. *Res. Sq.* rs-3 (2023).
28. Haze, T. et al. Influence on the accuracy in ChatGPT: Differences in the amount of information per medical field. *Int. J. Med. Inform.* **1**(180), 105283 (2023).
29. Alammari, D. M., Melebari, R. E., Alshaikh, J. A., Alotaibi, L. B., Basabeen, H. S., Saleh, A. F. & Abdulsalam, J. Beyond boundaries: The role of artificial intelligence in shaping the future careers of medical students in Saudi Arabia. *Cureus* **16**(9) (2024).

Author contributions

RM,RSAG,MA,AH,SA,RN,AS: Conceptualization, Writing—Original Draft. RSAG,NA,AH,RJ: Conceptualization, Study design, Data collection, Data Analysis, Writing—Original Draft, Critical Review of the Manuscript. MA,RSAG,NA: Data Collection, and Writing—Original Draft. RN,NA,SA: Conceptualization, Writing—Original Draft. AS: Data Collection, and Writing—Original Draft NA,AH,SA,RJ: Final review of the manuscript and Writing—Original Draft. All authors read and approved the final manuscript.

Funding

This work was supported by the Deanship of Scientific Research, Vice Presidency for Graduate Studies and Scientific Research, King Faisal University, Saudi Arabia [Grant No. KFU254749]

Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

The research ethics and review community of Altamash Institute of Dental Medicine has granted ethical clearance to carry out this research task. Ethics Review Committee reference number: AIDM/ERC/01/2024/02. The participants provided informed written consent to participate in the study.

Informed consent

Informed consent has been obtained from participants involved in this study.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-025-33704-1>.

Correspondence and requests for materials should be addressed to A.H. or R.J.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025